



SEMESTER END EXAMINATIONS – JANUARY 2020

Program	: B.E. : Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT- I

- Define an algorithm. Explain its various characteristics. CO1 (06)
 - Use KMP algorithm of pattern matching to find the pattern 'bac' in the following text abacbcab. Write the failure values and explain. CO1 (06)
 - Write an ADT for a sparse matrix and include necessary operations. CO1 (08)
- Illustrate how a two dimensional array is allocated dynamically, with a suitable example. CO1 (06)
 - Write necessary C functions to add two polynomials represented using an array. CO1 (08)
 - Write a recursive C function to find the length of a string. CO1 (06)

UNIT- II

- Implement the queue full and queue empty functions: CO2 (10)
 - For the circular queue.
 - For the non circular queue.
 - Write postfix form of the following expressions using stack in tabular form. CO2 (10)
 - $(d-c)/e+f/(g-k+l-i)$
 - $(b*b)-(4*k-l+(r-w)-h*o)$
- Write a C function that evaluates a **prefix** expression. CO2 (10)
(Hint: Use queue)
 - Write a C function for stackFull() for multiple stacks. If the i^{th} stack is full and the $(i+1)^{\text{th}}$ stack has free memory space, then increase the i^{th} stack memory space to store a new element, by acquiring the memory space from $(i+1)^{\text{th}}$ stack. CO2 (10)

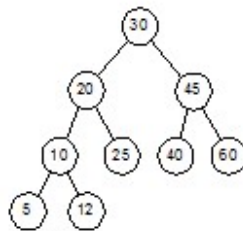
UNIT- III

- Write C functions for push() and pop() operations for a linked stack. CO3 (05)
 - Represent the following two polynomials using singly linked lists. CO3 (05)
$$A(x) = 5x^6 + 6x^4 - 4x + 10$$
$$B(x) = 4x^7 + 2x^3 + 7x^2 - 9$$
 - Write a C function to delete a node from a doubly linked circular list with a header node. Also illustrate with an example. CO3 (10)

6. a) Write a C function to multiply every alternate node by 5 in a singly linked list and display the entire list after modification. CO3 (06)
- b) Write a C function that searches for an integer NUM in a circular linked list. If NUM is in the list, function should return a pointer to that node. Otherwise it should return NULL. CO3 (08)
- c) Write a C function to append one doubly linked list to the end of another. CO3 (06)

UNIT- IV

7. a) Define each of the following with an example. CO4 (06)
 - i) Tree
 - ii) Degree of a tree
 - iii) Height of a tree.
- b) Construct the binary tree stepwise from its preorder (ABCD FEG) and in order (BAFDCEG) traversals. CO4 (06)
- c) Suppose that we have the following key values: 7, 16, 49, 82, 5, 31, 6, 2, and 44. Write out the MIN heap after each value is inserted into the heap. CO4 (08)
8. a) Define with an example each of the following: CO4 (06)
 - i) Binary tree
 - ii) Full binary tree
 - iii) Complete binary tree.
- b) CO4 (06)



Write the Pre-Order, In-Order and Post-Order traversals of the following binary tree.

- c) Explain the winner and loser trees with a suitable examples. CO4 (08)

UNIT- V

9. a) Demonstrate the construction of a multilists for a graph with an example. CO5 (08)
- b) Write a C function for depth first traversal over a graph. Show the depth first traversal of a graph with an example. CO5 (08)
- c) Justify the need for an AVL tree over unbalanced binary trees. CO5 (04)
10. a) Demonstrate the construction of adjacency matrix and adjacency lists for a graph. CO5 (08)
- b) What is breadth first traversal of a graph? Show it with an example. CO5 (06)
- c) Give example for winner tree and loser tree operations with ten runs. CO5 (06)
